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#include <iostream> #include <string> #include <vector> using namespace std;
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string gradeSchoolMult(string num1, string num2) {
    int n1 = num1.size();
    int n2 = num2.size();

    if (n1 == 0 || n2 == 0) {
        return "0";
    }

    vector<int> result(n1 + n2, 0);

    int i1 = 0, i2 = 0;

    for (int i = n1 - 1; i >= 0; i--) {
        int carry = 0;
        int n1_digit = num1[i] - '0';

        i2 = 0;

        for (int j = n2 - 1; j >= 0; j--) {
            int n2_digit = num2[j] - '0';

            int sum = n1_digit * n2_digit + result[i1 + i2] + carry;

            carry = sum / 10;
            result[i1 + i2] = sum % 10;

            i2++;
        }

        if (carry > 0) {
            result[i1 + i2] += carry;
        }

        i1++;
    }

    int i = result.size() - 1;
    while (i >= 0 && result[i] == 0) {
        i--;
    }

    if (i == -1) {
        return "0";
    }

    string str = "";

    while (i >= 0) {
        str += to_string(result[i--]);
    }

    return str; }

int main() {
    string num1 = "123456789";
    string num2 = "987654321";

    cout << "Number 1: " << num1 << endl;
    cout << "Number 2: " << num2 << endl;

    string product = gradeSchoolMult(num1, num2);

    cout << "Product: " << product << endl;
```

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return 0; }
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